

MPT Data-Driven Asset Allocation Model

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Markowitz’s Modern Portfolio Theory (MPT)

When deciding how to allocate an investment portfolio of multiple assets, the risk to return ratio is often not optimized. An unbalanced portfolio may assume too much risk in exchange for not enough returns. Harry Markowitz, an economist, developed the Modern Portfolio Theory (MPT) in the 1950’s. The MPT is simply a **statistical framework** or a foundation to determine the mathematically optimal weights for each asset in a portfolio.

These are the fundamental variables that are core to developing a basic MPT model.

Key Term	Definition & Application
Weight (w)	This is the variable denotation used to indicate the portion of the portfolio an asset is allocated, ranging between [0,1]
Expected Returns (μ)	This is the variable denotation used to indicate the expected returns (usually based on historical data) an asset is expected to gain.
Covariance Matrix (Σ)	A matrix displaying the covariance of the expected returns of an asset against another asset, shows how different assets move together or separately.
Risk Free Rate (r_f)	A rate of return through a highly stable instrument (i.e. a 10-year treasury) that is considered risk-free, part of the Sharpe ratio

These figures help us develop the **Sharpe ratio**, a formulaic way of representing the excess return (above the risk-free rate) that investing will give in exchange for taking on risk (volatility).

$$s = \frac{w^t \mu - r_f}{\sqrt{w^t \Sigma w}}$$

The numerator being the excess return of the weight exceeding the risk free rate and the denominator being the standard deviation (volatility) of the entire portfolio. The MPT framework applies optimization techniques to maximize the Sharpe ratio, maximizing the return given by taking more risk.

The Efficient Frontier

When comparing the risk to reward output out of thousands of portfolio combinations all with different weights, a pattern emerges. By investing in more **diverse** assets with non-matching correlations, the Sharpe ratio is significantly improved (that is to say, there is much more reward for taking on risk).

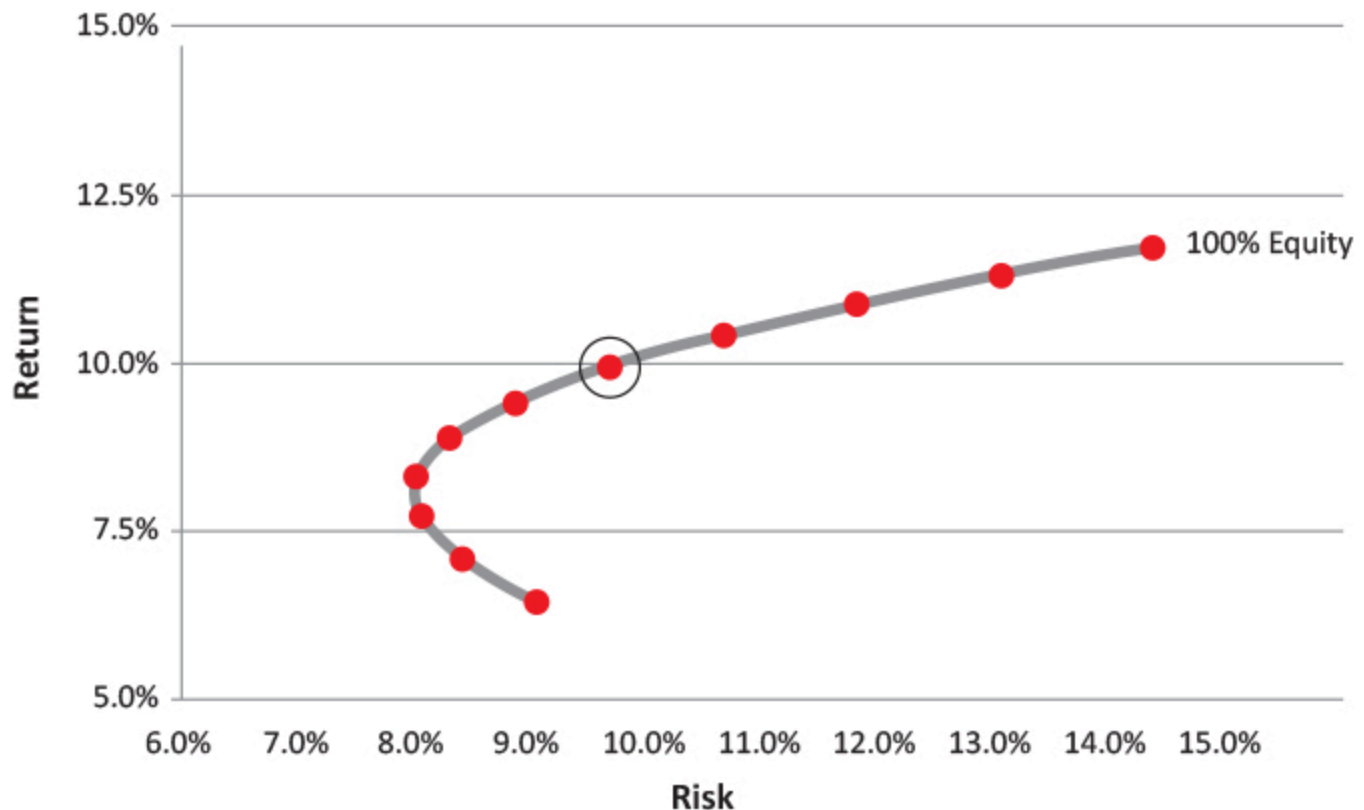


Fig.1

Fig.1 Credit. 'Efficient Frontier', Proactive Advisor Magazine, October 5, 2020

In reality, there are many efficient frontiers depending on the combination of assets used and the frontier continuously moves as time goes on. For the purposes of this educational project, we will use exclusively public equities.

Why does investing in **diverse** assets with **non-matching or negative correlations** hedge risk? If a portfolio includes two stocks with a correlation of $r = 0.5$, that means that ($r^2 = 0.25$) **25% of the variance are shared** by the two stocks, and **75% is not**. When one of those two stocks moves downwards, the other may counter or 'block out' that noise, and the portfolio gains more because of exposure to more industries and firms.

Let's put this example to play with real mathematics to demonstrate the point.

	Returns Over Period	Volatility Over Period
Stock A	10%	20%
Stock B	8%	16%
Stock C	5%	10%

Let's say that :

- Stock A and Stock B have a correlation of $r = 0.5$
- Stock B and Stock C have a correlation of $r = 0.25$
- Stock C and Stock A have a correlation of $r = -0.5$

That would create the following covariance matrix:

	Stock A	Stock B	Stock C
Stock A	20% * 20% = 0.04	0.5 * 20% * 16% = 0.016	-0.5 * 20% * 10% = -0.01
Stock B	0.5 * 20% * 16% = 0.016	16% * 16% = 0.0256	0.25 * 16% * 10% = 0.004
Stock C	-0.5 * 20% * 10% = -0.01	0.25 * 16% * 10% = 0.004	10% * 10% = 0.01

Since variance and covariance are a squared translation (and thus represent) of the volatility of a stock, we can see the benefit of combining different stocks, and diversifying will decrease the volatility of the portfolio. We can see that a combination of stock A and stock C reduces the risk by far compared to other options, so let's say that a portfolio is built with the following weights of stocks A, B, and C:

- Stock A with a weight of 33%
- Stock B with a weight of 0%
- Stock C with a weight of 67%

Now we can determine our overall portfolio volatility:

$$\sigma = \sqrt{[0.33, 0, 0.67] \begin{bmatrix} 0.04 & 0.016 & -0.01 \\ 0.016 & 0.0256 & 0.004 \\ -0.01 & 0.004 & 0.01 \end{bmatrix} \begin{bmatrix} 0.33 \\ 0 \\ 0.67 \end{bmatrix}} = 0.0665$$

And our portfolio expected returns would be as follows:

$$E(x) = [0.33, 0, 0.67] \begin{bmatrix} 0.1 \\ 0.08 \\ 0.05 \end{bmatrix} = 0.0665$$

Meaning that compared to directly investing in one stock, this portfolio would return:

	Returns Over Period	Volatility Over Period
100% Stock A	10%	20%
100% Stock B	8%	16%
100% Stock C	5%	10%
Portfolio: 33% Stock A 67% Stock C	6.7% (Higher returns than a 100% Stock C investment)	6.7% (Volatility lower than any individual investment)

This is the foundation of the efficient frontier. By properly diversifying a portfolio, **risk (volatility) decreases, and returns increase**, as seen in the mock portfolio (See Fig.1 for Visual). The question now arises, when dealing with a universe of 10, 20, 50, 100 different investment options (whether that be stocks, bonds, alternative funds), **what weights are proper** for each option to maximize the sharpe ratio of risk to return?

There are many ways to go about determining proper weights, and it all comes down to the data used. Many firms will develop complex models and multiple layers of pre and post processing before determining which assets to buy, when to buy them, and how much of each is optimal.

Sequential Least Squares Quadratic Programming (SLSQP)

Sequential Least Squares Quadratic Programming is an **iterative** (gets more accurate each time it runs) **optimization** algorithm designed to solve **complex nonlinear problems** (like a large set of profile weights that have extremely complex functions and relations in relation to other weights) according to limitations.

At the centre of SLSQP is the Lagrangian Function, the target function accounting for necessary limitations:

$$L(x, \lambda, \mu) = f(x) - \sum \lambda_i h_i(x) - \sum \mu_j g_j(x)$$

Key Term	Definition & Application
$f(x)$	This is the objective function , it represents what the optimizer is trying to solve (minimize/maximize) in this case, it is the Sharpe ratio as a function of a matrix, x, of all the weights in the portfolio
$\sum \lambda_i h_i(x)$	This is the formula for the equality constraints, these are fixed (rigid) constraints that are penalized by a multiplier (lambda) if broken. An example of this is that all portfolio weights added up must equal 1.
$\sum \mu_j g_j(x)$	This is the formula for the inequality constraints, these are ranged (inequality) constraints that are penalized by a multiplier (mu) if broken. An example of this is that the weight of a stock must be greater than or equal to 0.

Important Note: Penalty Constraints

There is a good reason why we use weighted penalties instead of hard stops. For example, although we prefer to have positive weights for each stock, technically, a negative weight can be executed by shorting the stock but investors like to avoid it. By penalizing the model but not preventing it, investors can attribute their own cost multiplier representing the extra cost of breaking the rule.

The Lagrangian function is a good start, however issues arise when trying to put it in practice. The biggest issue that the Lagrangian function encounters is that the objective function, $f(x)$, is extraordinarily complex (the Sharpe ratio formula for **hundreds of stocks** must account for **hundreds of thousands of covariances**, taking the square root of a **variance requiring millions of calculations**, etcetera).

To simplify the objective function, $f(x)$, SLSQP uses **quadratic approximation** using a second-order Taylor expansion. Instead of mapping out the entire objective function the algorithm creates a simple bowl (quadratic) approximation that represents its immediate surroundings to determine which direction optimizes the objective further. Once it decides the next direction, it changes its position and recreates a **new quadratic approximation** at that location.

This is the minimization problem of the quadratic approximation that the model solves each iteration:

$$\min_d \frac{1}{2} d^t H d + \nabla f(x_k) d$$

To visualize this, let's analyze how two stocks would interact; stocks 1 and 2. Instead of minimizing the negative sharpe ratio (which in-practice is the same by) we will minimize risk (variance $\propto$ volatility) for simplicity.

	Returns Over Period t	Volatility Over Period t
Stock 1	5%	5%
Stock 2	5%	5%

The two are correlated by a factor of $r = 0.5$ to produce the following Covariance Matrix:

	Stock 1	Stock 2
Stock 1	5% * 5% = 0.0025	0.5 * 5% * 5% = 0.00125
Stock 2	0.5 * 5% * 5% = 0.00125	5% * 5% = 0.0025

The model uses the second-order Taylor expansion structure to approximate the function with the gradient and hessian (the minimization function seen above).

The gradient is a representation of the first-order derivative of a multivariable problem, the hessian the second-order derivative, and looks like this:

$$w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}, \nabla f(w) = \begin{bmatrix} f_{w_1} \\ f_{w_2} \end{bmatrix}, H = \begin{bmatrix} f_{w_1 w_1} & f_{w_1 w_2} \\ f_{w_2 w_1} & f_{w_2 w_2} \end{bmatrix}$$

The approximation problem would look like this:

$$\min_d \frac{1}{2} [\Delta w_1, \Delta w_2] \begin{bmatrix} f_{w_1 w_1} & f_{w_1 w_2} \\ f_{w_2 w_1} & f_{w_2 w_2} \end{bmatrix} \begin{bmatrix} \Delta w_1 \\ \Delta w_2 \end{bmatrix} + [f_{w_1}, f_{w_2}] \begin{bmatrix} \Delta w_1 \\ \Delta w_2 \end{bmatrix}$$

Let's say that the initial guess for weights is to buy 100% of stock 1 and 0% of stock 2:

$$w = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

The approximation problem now becomes:

$$\min_d \frac{1}{2} d^t \begin{bmatrix} 0.005 & 0.0025 \\ 0.0025 & 0.005 \end{bmatrix} d + \left(\begin{bmatrix} 0.005 & 0.0025 \\ 0.0025 & 0.005 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)^t$$

Which becomes:

$$\min_d \frac{1}{2} d^t \begin{bmatrix} 0.005 & 0.0025 \\ 0.0025 & 0.005 \end{bmatrix} d + [0.005, 0.0025] d$$

Important Note: Hessian, Covariances, and Negative Sharpe

The Hessian Matrix and Covariance Matrix are two distinct but related concepts as they both measure the change of variables in respect with others. In this case, the Hessian Matrix can be derived to two times the Covariance Matrix but this may not be the case for other calculations/models.

In this case, we are minimizing risk, so the hessian matrix and guess can be multiplied to derive the true gradient. However, when minimizing the negative sharpe ratio, the algorithm takes the first-order derivative of the target function and constructs an approximate Hessian matrix via the BFGS method instead to build the minimization problem.

From this, the model optimizes for the direction d (represented as a change of magnitude x that must equal out to 0 so that both weights can follow the equality constraint where the both weights must add up to 1, thus changes in weights must be inversely proportional to each other) and simplifies the problem.

$$\min_d \frac{1}{2} [x, -x] \begin{bmatrix} 0.005 & 0.0025 \\ 0.0025 & 0.005 \end{bmatrix} \begin{bmatrix} x \\ -x \end{bmatrix} + [0.005, 0.0025] \begin{bmatrix} x \\ -x \end{bmatrix}$$

Which becomes:

$$\min_x 0.0075x^2 + 0.0075x$$

And X (and the direction vector) is solved as:

$$x = -0.5 \qquad d = \begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix}$$

Thus, the model determines that the direction for the next step must be to change to weigh stock 2 heavier and stock 1 lighter. It takes a step in that direction and reconstructs the minimization problem.

This process repeats until the model reaches a balanced solution that minimizes risk.

$$w = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$$